STAT6611

- Lecture 17 (R, Oct 22)
- solutions posted to Homework 3
- Take home midterm due F, Oct 23 (midnight in Canvas)
- No new homework this week
  (next assignment next R, Oct 29)
  Homework 4

3.4 Distributions of Quadratic Forms

For any quadratic form $\underline{x}'A\underline{x}$, we have

$$\underline{x}'A\underline{x} = \left[\underline{x}'A\underline{x} + \left(\underline{x}'A\underline{x}\right)'\right]/2 = \left(\underline{x}'A\underline{x} + \underline{x}'A'\underline{x}\right)/2 = \underline{x}'\left[(A+A')/2\right]\underline{x} = \underline{x}'A_1\underline{x},$$

where $A_1 = (A+A')/2$ is symmetric. Thus, without loss of generality, in our discussion of distributions of quadratic forms, **we assume that the matrix A associated with any quadratic form $\underline{x}'A\underline{x}$ is symmetric.**

Theorem 3.14. *If $\underline{x} \sim N(\underline{\mu}, \Sigma)$, then a sufficient condition for a quadratic form $\underline{x}'A\underline{x}$ and a vector $B\underline{x}$ of linear forms to be independent is that $B\Sigma A = 0$.*

Corollary. If $\underline{x} \sim N(\underline{\mu}, \Sigma)$, then a sufficient condition for the quadratic forms $\underline{x}'A\underline{x}$ and $\underline{x}'B\underline{x}$ to be independent is that $B\Sigma A = 0$.

Above if $\underline{x} \sim N(\underline{\mu}, \Sigma)$ & Σ is invertible, then $\underline{x}'A\underline{x}$ & $\underline{x}'B\underline{x}$ are independent $\Leftrightarrow B\Sigma A = 0$

Remarks. If Σ is nonsingular, it can be shown that the condition $B\Sigma A = 0$ in Theorem 3.14 and the corollary to Theorem 3.15 is necessary as well as sufficient. The proof of necessity is much more difficult than that of sufficiency, see the paper by Harville and Kempthorne, "An alternative way to establish the necessity part of the classical result on the statistical independence of quadratic forms" (*Linear Algebra and Its Applications*, Volume 264, 1997, 205–215). If Σ is singular, the condition $B\Sigma A = 0$ in Theorem 3.14 and the corollary to Theorem 3.15 is no longer necessary.

← a generalization of past corollary

Theorem 3.15. *If $\underline{x} \sim N(\underline{\mu}, \Sigma)$, then a sufficient condition for the quadratic forms $\underline{x}'A_1\underline{x}$, ..., $\underline{x}'A_k\underline{x}$ to be mutually independent is that $A_i\Sigma A_j = 0$ for $i = 1, \ldots, k$, $j = 1, \ldots, k$, and $i \neq j$.*

↑ ↑ each pair of A_i & A_j $i \neq j$

Proof. By Theorem 3.5,

Same proof as before

$$\begin{bmatrix} A_1\underline{x} \\ \vdots \\ A_k\underline{x} \end{bmatrix} = \begin{bmatrix} A_1 \\ \vdots \\ A_k \end{bmatrix} \underline{x} \sim \text{MVN}$$

has a multivariate normal distribution. If $A_i\Sigma A_j = 0$ for $i = 1, \ldots, k$, $j = 1, \ldots, k$, and $i \neq j$, then $\text{cov}(A_i\underline{x}, A_j\underline{x}) = A_i\Sigma A_j' = A_i\Sigma A_j = 0$ for all $i \neq j$. Then, by Theorem 3.7, $A_1\underline{x}, \ldots, A_k\underline{x}$ are mutually independent. This implies that $\underline{x}'A_1\underline{x}, \ldots, \underline{x}'A_k\underline{x}$ are mutually independent, because for $i = 1, \ldots, k$,

$$\underline{x}'A_i\underline{x} = \underline{x}'A_iA_i^-A_i\underline{x} = (A_i'\underline{x})'A_i^-(A_i\underline{x}) = (A_i\underline{x})'A_i^-(A_i\underline{x})$$

is a function of $A_i\underline{x}$. □

Lemma 3.2. *An $n \times n$ symmetric matrix A is idempotent of rank r if and only if it has the characteristic root 1 with multiplicity r and the characteristic root 0 with multiplicity $n - r$.*

In other words, A symmetric/idempotent ($A^2 = A$) $\Leftrightarrow A = P \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} P'$ orthogonal P ($P'P = I_n$)

Proof. Corresponding to an $n \times n$ symmetric matrix A of rank r, there exists an orthogonal matrix P such that

$$P'AP = D,$$

i.e. $A = P \begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix} P'$ holds for D_r with non-zero entries

where D is a diagonal matrix whose diagonal elements $d_1, \ldots, d_n$ are the characteristic roots of A. Necessarily, $\text{rank}(D) = r$, that is, D has r nonzero elements.

A is idempotent $\Leftrightarrow A^2 = A$

$\Leftrightarrow P \begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix} P'P \begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix} P' = P \begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix} P'$ ($P'P = I$)

$\Leftrightarrow \begin{bmatrix} D_r^2 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix} \Leftrightarrow D_r^2 = D_r$ (D_r^{-1} exists)

$\Leftrightarrow D_r = I_r$

Lemma 3.3. *Take $r_1(x), r_2(x), s_1(x)$, and $s_2(x)$ to be polynomials in a real variable x. Suppose that*

$$r_2(x) \equiv \gamma(x - \lambda_1)^{m_1}(x - \lambda_2)^{m_2} \cdots (x - \lambda_k)^{m_k},$$

where k is a nonnegative integer, $m_1, \ldots, m_k$ are positive integers, $\lambda_1, \ldots, \lambda_k$ are constants, and γ is a nonzero constant. Suppose further that

$$\frac{s_1(x)}{s_2(x)} = \exp\left[\frac{r_1(x)}{r_2(x)}\right] \quad \text{for} \quad x \in J,$$

where J is some nondegenerate interval that does not include $\lambda_1, \ldots, \lambda_k$ or any roots of $s_2(x)$. Then, there exist constants α and β such that $r_1(x) = \alpha r_2(x)$ and $s_1(x) = \beta s_2(x)$ for all x.

The proof of Lemma 3.3 is discussed in the paper by Driscoll and Gundberg, "A history of the development of Craig's theorem" (*The American Statistician*, Volume 40, 1986, 65–70).

ie Σ^{-1} exists

Theorem 3.16. *Let* $\underline{x} \sim N(\underline{\mu}, \Sigma)$, *where* $\underline{x}$ *is an* $r \times 1$ *vector and* $\text{rank}(\Sigma) = r$. *Let* A *be an* $r \times r$ *symmetric matrix of constants. If* $A\Sigma$ *is idempotent of rank* $s \geq 1$, *then* $\underline{x}'A\underline{x} \sim \chi^2\left(s, \frac{1}{2}\underline{\mu}'A\underline{\mu}\right)$. *Conversely, if* $\underline{x}'A\underline{x} \sim \chi^2(s, \lambda)$ *for some* λ, *then* $A\Sigma$ *is idempotent of rank* s.

this characterizes what is needed for a quadratic form $\underline{x}'A\underline{x}$ to chi-square distributed (possibly non-central)
We'll later see that "Σ" must be a generalized inverse of A (i.e. $A\Sigma A = A$). Also, note $s = \text{rank}(A\Sigma) = \text{rank}(A) \leq r$.

We first prove the theorem for the case $\Sigma = I$, which simplifies as follows.

Let $\underline{x} \sim N(\underline{\mu}, I_r)$. *Let* A *be an* $r \times r$ *symmetric matrix of constants. If* A *is idempotent of rank* $s \geq 1$, *then* $\underline{x}'A\underline{x} \sim \chi^2\left(s, \frac{1}{2}\underline{\mu}'A\underline{\mu}\right)$. *Conversely, if* $\underline{x}'A\underline{x} \sim \chi^2(s, \lambda)$ *for some* λ, *then* A *is idempotent of rank* s.

Proof. Suppose first $r \times r$ A is idempotent of rank r.

$\Rightarrow$ If $r \times r$ A has rank$(A) = r$ & A is idempotent ($A^2 = A$)
$\Rightarrow$ since A^{-1} invertible, $A = I_r$
Then, $\underline{x}'A\underline{x} = \underline{x}'\underline{x} \sim \chi^2(r, \frac{1}{2}\underline{\mu}'\underline{\mu})$ ($\underline{x} \sim N(\underline{\mu}, I_r)$; $\frac{1}{2}\underline{\mu}'A\underline{\mu}$)

If A is idempotent of rank s where $1 \leq s < r$, then by Lemma 3.2, there exists an orthogonal matrix $P = (P_1, P_2)$, where P_1 is an $r \times s$ matrix, such that

$$A = P \begin{bmatrix} I_s & 0 \\ 0 & 0 \end{bmatrix} P' = (P_1, P_2) \begin{bmatrix} I_s & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} P_1' \\ P_2' \end{bmatrix} = (P_1, 0) \begin{bmatrix} P_1' \\ P_2' \end{bmatrix} = P_1 P_1'.$$

(A: $r \times r$; P: $r \times s$; P_1: $r \times s$; P_1': $s \times r$; $P_1 P_1'$: $r \times s$ $s \times r$)

Lemma 3.2

$P'P = I_r$ P $r \times r$

Note that

$$\begin{bmatrix} I_s & 0 \\ 0 & I_{r-s} \end{bmatrix} = I_r = P'P = \begin{bmatrix} P_1' \\ P_2' \end{bmatrix} (P_1, P_2) = \begin{bmatrix} P_1'P_1 & P_1'P_2 \\ P_2'P_1 & P_2'P_2 \end{bmatrix},$$

which implies that $P_1'P_1 = I_s$. Let $\underline{z} = P_1'\underline{x}$. Then, $\underline{x}'A\underline{x} = \underline{x}'P_1P_1'\underline{x} = \underline{z}'\underline{z}$.

Next suppose (still assuming $\underline{x} \sim N(\underline{\mu}, I_r)$) that $\underline{x}'A\underline{x} \sim \chi^2(s, \lambda)$ for some λ. Need to show A is idempotent of rank s.

Let $m = \text{rank}(A)$. Since A is symmetric, for an orthogonal matrix $P = (P_1, P_2)$, where P_1 is an $r \times m$ matrix, we can write

$$A = P \begin{bmatrix} D_m & 0 \\ 0 & 0 \end{bmatrix} P' = (P_1, P_2) \begin{bmatrix} D_m & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} P_1' \\ P_2' \end{bmatrix} = P_1 D_m P_1'.$$

where D_m is a diagonal matrix with entries (say) $d_1, \ldots, d_m$.

Since $P_1'P_1 = I_m$ and since $\underline{x} \sim N(\underline{\mu}, I_r)$, we have $\underline{z} = P_1'\underline{x} \sim N(P_1'\underline{\mu}, I_m)$. Note

$$\underline{x}'A\underline{x} = \underline{x}'P_1 D_m P_1'\underline{x} = \underline{z}'D_m\underline{z} = \sum_{i=1}^{m} d_i z_i^2, \quad \underline{z} = \begin{pmatrix} z_1 \\ \vdots \\ z_m \end{pmatrix}.$$

$$M_{\underline{x}'A\underline{x}}(t) = \mathrm{E}\exp\left(t\underline{x}'A\underline{x}\right) = \mathrm{E}\exp\left(t\sum_{i=1}^{m} d_i z_i^2\right) = \mathrm{E}\prod_{i=1}^{m}\exp\left(td_i z_i^2\right)$$
$$= \prod_{i=1}^{m}\mathrm{E}\exp\left(td_i z_i^2\right) = \prod_{i=1}^{m} M_{z_i^2}(td_i)$$
$$= \prod_{i=1}^{m}\left\{(1-2d_i t)^{-1/2}\exp\left(\frac{2td_i\frac{1}{2}\mu_{z_i}^2}{1-2d_i t}\right)\right\}$$

provided t satisfies $td_i < 1/2$ for each $i = 1, \ldots, m$.

By assumption, though, $\underline{x}'A\underline{x} \sim \chi^2(s, \lambda)$ for some λ, so it must be

$$M_{\underline{x}'A\underline{x}}(t) = \left\{(1-2t)^{-s/2}\exp\left(\frac{2t\lambda}{1-2t}\right)\right\}, \quad t < 1/2.$$

So for small t around 0 (where $|t| < 1/2$ and $|t|\min_{1\le i\le m}|d_i| < 1/2$), we have

$$\left\{(1-2t)^{-s/2}\exp\left(\frac{2t\lambda}{1-2t}\right)\right\} = \prod_{i=1}^{m}\left\{(1-2d_i t)^{-1/2}\exp\left(\frac{2td_i\frac{1}{2}\mu_{z_i}^2}{1-2d_i t}\right)\right\},$$

which can be re-arranged algebraically as

$$\frac{\prod_{i=1}^{m}(1-2d_i t)}{(1-2t)^s} = \exp\left(\frac{-4t\lambda}{1-2t} + 4t\sum_{i=1}^{m}\frac{d_i\frac{1}{2}\mu_{z_i}^2}{1-2d_i t}\right).$$

By Lemma 3.1, a constant β exists where $\prod_{i=1}^{m}(1-2d_it) = \beta(1-2t)^s$ for all $t \in \mathbb{R}$.

For $t=0$, $1 = \beta \cdot 1$ or $\beta = 1$

It must be that $m=s$ (degrees match)

& roots of RHS (roots at $t=\frac{1}{2}$ always)

= roots of LHS (roots at $t=\frac{1}{2d_i}$)

$\Rightarrow d_i = 1, i=1,\ldots,s=m$

We now prove the theorem for the case Σ is nonsingular (by assumption) but $\Sigma \neq I$.

Express $\Sigma = \Gamma'\Gamma$, where Γ is a nonsingular matrix, and let $\underline{w} = \left(\Gamma'\right)^{-1}\underline{x}$. Then,

$$\underline{x}'A\underline{x} = \underline{w}'\Gamma A\Gamma'\underline{w} \quad \text{and} \quad \underline{w} \sim N\left[\left(\Gamma'\right)^{-1}\underline{\mu}, I\right].$$

Thus, we can apply the theorem for the case $\Sigma = I$ already proved above to $\underline{w}'\left(\Gamma A\Gamma'\right)\underline{w}$. That is, if $\Gamma A\Gamma'$ is idempotent of rank $s \geq 1$ (note that $\Gamma A\Gamma'$ is symmetric), then

$$\underline{w}'\left(\Gamma A\Gamma'\right)\underline{w} \sim \chi^2\left[s, \frac{1}{2}\underline{\mu}'\Gamma^{-1}\left(\Gamma A\Gamma'\right)\left(\Gamma'\right)^{-1}\underline{\mu}\right], \text{ that is, } \underline{x}'A\underline{x} \sim \chi^2\left(s, \frac{1}{2}\underline{\mu}'A\underline{\mu}\right);$$

and, conversely, if $\underline{x}'A\underline{x} \sim \chi^2(s,\lambda)$ for some λ, that is, if $\underline{w}'\left(\Gamma A\Gamma'\right)\underline{w} \sim \chi^2(s,\lambda)$ for some λ, then $\Gamma A\Gamma'$ is idempotent of rank s.

It remains to show that $\Gamma A\Gamma'$ is idempotent of rank s if and only if $A\Sigma$ is idempotent of rank s.